We have shown $P(t) = \Psi(t) e^{-Rt}$, where

$\Psi(t)$ is any fund'l matrix for $A(t)$

R is any constant matrix

Transforms $\dot{\underline{x}}(t) = A(t)\underline{x}(t) \Rightarrow \dot{\underline{z}}(t) = R\underline{z}(t)$

so hence

$$\Phi(t,\tau) = P(t) e^{R(t-\tau)} P^{-1}(\tau)$$

STM for $\underline{x}$ — STM for $\underline{z}$

Note if $R = \text{diag}\{r_i\}$ then $e^{Rt} = \text{diag}\{e^{r_i t}\}$

$\Rightarrow$ Whether this gives useful insight depends on required $P(t)$. Generally want to ensure (in addition to nonsingularity) that

- $P(t) \not\to 0$
- $P^{-1}(t) \not\to 0$

As $t \to \infty$

A transformation with these add'l restrictions is known as a "Lyapunov transformation"

Challenge is to ensure $P(t)$ has this property w/o solving for $\Psi(t)$ directly.

Linear time-periodic system (LTP)

$\exists\, T < \infty$ s: $A(t+T) = A(t)$ $\forall t \geq t_0$

For such systems, $P(t)$ itself is actually also periodic, and hence a Lyap xform:

$\Rightarrow$

Let $Q = \Phi(t_0+T, t_0)$ "monodromy matrix"

Define R by $e^{RT} = Q$

$$\Rightarrow R = \left(\tfrac{1}{T}\right) \log Q$$

matrix log

MatLab "logm"

Define

$$P(t) = \Phi(t, t_0) e^{-Rt}$$

Claim: $P(t+T) = P(t)$ $\forall t \geq t_0$

$$P(t+T) = \Phi(t+T, t_0) e^{-R(t+T)}$$

$$= \Phi(t+T, t_0) e^{-RT} e^{-Rt}$$

$e^{-RT} = \left[e^{RT}\right]^{-1} = Q^{-1} =$

$$e^{-RT} = \left[e^{RT}\right]^{-1} = Q^{-1} =$$

$$= \left[\Phi(t_0+T, t_0)\right]^{-1}$$

$$= \Phi(t_0, t_0+T)$$

So now

$$P(t+T) = \Phi(t+T, t_0)\Phi(t_0, t_0+T)e^{-Rt}$$

$$= \Phi(t+T, t_0+T)e^{-Rt}$$

$$\overset{?}{=} \Phi(t, t_0)e^{-Rt} = P(t)$$

only if $\Phi(t, t_0) = \Phi(t+T, t_0+T)$

Is this true?

Let $\Phi_0(t) = \Phi(t, t_0)$

$\Phi_T(t) = \Phi(t+T, t_0+T)$

$\dot{\Phi}_0(t) = \partial/\partial t\, \Phi(t, t_0) = A(t)\Phi_0(t)$, $\Phi_0(t_0) = \mathrm{II}$

$\dot{\Phi}_T(t) = \partial/\partial t\, \Phi(t+T, t_0+T) = A(t+T)\Phi_T(t)$

$= A(t)\Phi_T(t)$, $\Phi_T(t_0) = \mathrm{II}$

By uniqueness of sol'ns for linear DES, $\Phi_0(t) = \Phi_T(t)$

hence

$$\Phi(t+T, t_0+T) = \Phi(t, t_0)$$

— true for time-periodic systems

and $P(t+T) = P(t)$ for this $P(t)$

=

So $P(t)$ provides a Lyapnov transformation turning an LTP system into the LTI system

$$\dot{z} = Rz \qquad R = \left(\tfrac{1}{T}\right)\log_m Q$$

and

$$\Phi(t, \tau) = P(t)e^{R(t-\tau)}P^{-1}(\tau)$$

$P(t)$ periodic

"Floquet transformation"

splits analysis into LTI part and periodic part.

Stability and convergence of homog. sol'ns

$$\underline{x}(t) = \Phi(t,t_0)\underline{x}_0$$

= In simple LTI/S/SO

Cond'n is $y_h(t) \to 0$ for all ICs

$\Leftrightarrow$ all poles in LHP.

To generalize such ideas to vector f'ns, we'll use vector norms: $\|\underline{x}\|$ ~ a measure of the geometric length of a vector

Many different metrics possible, all must satisfy

- $\|\underline{x}\| \geq 0$
- $\|\underline{x}\| = 0 \Leftrightarrow \underline{x} = 0$
- $\|\alpha \underline{x}\| = |\alpha| \, \|\underline{x}\|$
- $\|\underline{x} + \underline{y}\| \leq \|\underline{x}\| + \|\underline{y}\|$ (Triangle ineq.)

Euclidean / 2-norm

- $\|\underline{x}\|_2 = \left[\sum_{k=1}^{n} x_k^2\right]^{1/2}$

also common are

1-norm: $\|\underline{x}\|_1 = \sum_{k=1}^{n} |x_k|$

∞-norm: $\|\underline{x}\|_\infty = \max_{1 \leq k \leq n} |x_k|$

p-norm: $\|\underline{x}\|_p = \left[\sum_{k=1}^{n} |x_k|^p\right]^{1/p}$

p integer ≥ 1

= Norm equivalence thm:

for every choice of p, q (integers ≥ 1) there exists $c_{pq} > 0$ and $\bar{c}_{pq} < \infty$ so that

$$c_{pq} \|\underline{x}\|_q \leq \|\underline{x}\|_p \leq \bar{c}_{pq} \|\underline{x}\|_q .$$

= Matrix induced norms:

$$\|A\|_i \triangleq \max_{\|\underline{x}\| \neq 0} \frac{\|A\underline{x}\|}{\|\underline{x}\|}$$

$\Rightarrow \|A\underline{x}\| \leq \|A\|_i \, \|\underline{x}\|$

Calculation of $\|A\|_i$ depends on choice of vector norm

$$\|A\|_{i,2} = \sqrt{\lambda_{max}(A^TA)}$$

$$\|A\|_{i,1} = \max_j \sum_{i=1}^{n} |a_{ij}| \quad \text{(max abs col sum)}$$

$$\|A\|_{i,\infty} = \max_i \sum_{j=1}^{n} |a_{ij}| \quad \text{(max abs row sum)}$$

Any induced norm satisfies:

- $\|A+B\|_i \le \|A\|_i + \|B\|_i$
- $\|AB\|_i \le \|A\|_i \|B\|_i$

=

Use these notions to describe stability/Bddness/convergence, first of state eq'n, then of Input/output Behavior

=

Example: $\exists B < \infty$ such that

$$\|\underline{x}(t)\| \le B < \infty \quad \forall t \ge t_0$$

or

$$\|\underline{x}(t)\| \to 0 \text{ as } t \to \infty.$$

=

More formally, a system is stable i.s.L. // — "in the sense of Lyapunov."

if $\exists \gamma < \infty$ such that

$$\|\underline{x}(t)\| \le \gamma \|\underline{x}_0\| \quad \forall t \ge t_0$$

Note

$$\|\underline{x}(t)\| = \|\Phi(t,t_0)\underline{x}_0\|$$

$$\le \|\Phi(t,t_0)\|_i \|\underline{x}_0\|$$

$\Rightarrow$ system is stable (ish) if

$$\|\Phi(t,t_0)\|_i \le \gamma < \infty \quad \forall t.$$

i.e. $\|\Phi(t,t_0)\|_i$ is Bdd.

$\Rightarrow$ The system is <u>asymptotically stable</u> if $\|\underline{x}(t)\| \to 0$ as $t \to \infty$.

$\Rightarrow \|\Phi(t,t_0)\|_i \to 0$ as $t \to \infty$.

How to deduce if Φ has these properties if we don't know exactly what Φ looks like?

We could equivalently think in terms of the change in $\|\underline{x}(t)\|$ over time,

i.e. $\frac{d}{dt}\|\underline{x}(t)\| \leq 0$

$\Rightarrow$ stable i.s.L { in fact, $\omega/\delta = 1$, so perhaps more restrictive than necessary...

or

$$\frac{d}{dt}\|\underline{x}(t)\|_2^2 \leq 0$$

$$\frac{d}{dt}(\underline{x}^T\underline{x}) = \left(\frac{d}{dt}\underline{x}^T\right)\underline{x} + \underline{x}^T\left(\frac{d}{dt}\underline{x}\right)$$

w/ homog. dyn $\dot{\underline{x}} = A\underline{x}$

$$\frac{d}{dt}\|\underline{x}\|_2^2 = (A\underline{x})^T\underline{x} + \underline{x}^T(A\underline{x})$$

$$= \underline{x}^T[A^T + A]\underline{x}$$ { "quadratic form"

≤ 0 ?? - must investigate!